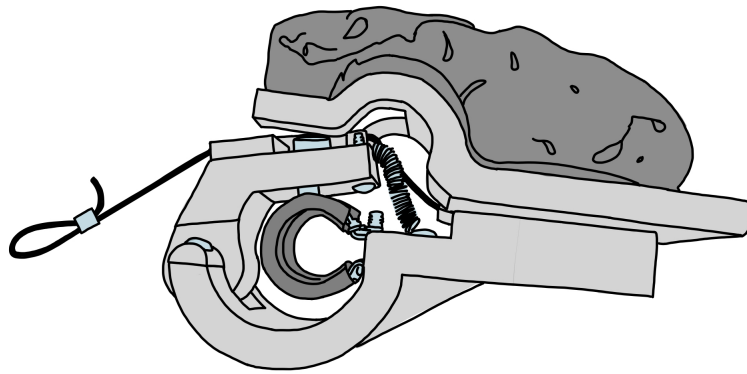


VeloBrake Assist

User Manual

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Group: P6, Adapted environment for ectrodactily

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1. Introduction

VeloBrake Assist was designed to help cyclists with reduced hand function ride more safely and confidently. By providing a secure and comfortable hand grip on the handlebar together with an intuitive braking mechanism, the device enables users to maintain control of their bike without having to completely relearn how to ride.

Compatible with most bicycles featuring standard-sized handlebars, VeloBrake Assist attaches securely to the handlebar once the original grip coating is removed. The system can also be mounted and removed, allowing it to be transferred between different bicycles rather than remaining permanently fixed to a single bike.

Braking is achieved through a simple downward wrist movement applied to a custom-molded hand support. As pressure is applied, a cable mechanically activates the bike brake, providing smooth and modulable braking control. While some practice may be required at first, the device is designed to feel as natural and intuitive as possible, allowing the user to focus on the ride rather than the mechanism itself.

2. The Device

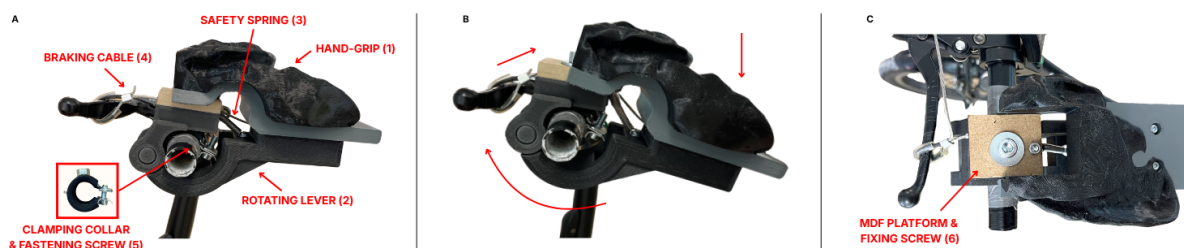


Figure 1: VeloBrake Assist. A. Components; B. Activation movement; C. MDF platform & fixing screw.

Hand Grip (1)

The hand grip is the part on which the user rests their hand during rides. It was designed from a 3D scan of a custom mold to ensure the best possible fit and comfort. The custom hand grip is partially manufactured in TPU (in black) to improve comfort and flexibility, while the remaining structure is made of PETG.

Rotating Lever (2)

The rotating lever is the moving part of the mechanism. It is made of PET-CF and anchors the braking cable and the safety spring. When the user pushes downward with their wrist, the rotating lever rotates and pulls the braking cable, activating the bicycle

brake.

Safety Spring (3)

The safety spring provides resistance to the mechanism in order to reduce the risk of accidental braking. Its role is to keep the rotating lever in its resting position until sufficient pressure is intentionally applied.

Braking Cable (4)

The braking cable is a 2 mm steel cable that connects the rotating lever to the bicycle brake. The cable is attached to the rotating lever and the brake using cable grippers. When the mechanism is activated, the cable is pulled through the guide canal and activates the brake.

Attachment Mechanism: Clamping Collar, Fastening & Fixing Screw (5)

The attachment mechanism is a steel clamping collar internally coated with rubber to improve grip and stability on the handlebar. Its ends are tightened together via a fastening screw, securing it onto the handlebar. The fixed part of the braking mechanism connects to the clamping collar via a fixing screw, forming a rigid and stable assembly once tightened.

3. How to Mount & Use the Device

3.1. Mounting the Device on the Bike

1. Remove the original grips or coatings from both sides of the bicycle handlebar.
2. Position the clamping collar on the handlebar and tighten the fastening screw using a flathead screwdriver until firmly secured.
3. Screw the rest of the device onto the clamping collar using an Allen key to tighten the fixing screw.
4. Pass the braking cable around the existing brake lever on the handlebar and secure it fastening a 2mm steel cable gripper with pliers. If the cable tension needs to be adjusted, remove the cable gripper with pliers, reposition the cable to the desired length, and secure it with a new 2mm steel cable gripper.

3.2. Using the Device

1. Place the left hand comfortably on the hand grip and the right hand on the handlebar. Maintain a secure hold with both hands while riding.
2. To brake, push downward with the left wrist to activate the mechanism, and pull the brake lever with the right hand.

4. Precautions and Safety

To ensure safe and reliable use of VeloBrake Assist, please carefully follow the recommendations below before and during use.

- The device is recommended for use on the left hand only. The right hand should grip the handlebar directly to maintain control in case of mechanism failure and to ensure greater stability.
- The device is manufactured using 3D-printed PET - CF components. While the material offers good mechanical resistance, it may be damaged by strong impacts or prolonged exposure to high temperatures. Avoid leaving the device in excessively hot environments or under direct sunlight for extended periods.
- Before each use, verify that the device is securely attached to the handlebar. Apply moderate force to check for any movement or instability. If the device shifts or rotates, tighten the fastening screw before riding.
- Regularly inspect the brake cable to ensure it is correctly tensioned and securely connected. A cable that is too loose may disengage from the brake mechanism during use, creating a safety hazard.
- VeloBrake Assist is intended for use in relatively controlled riding environments such as city streets, bike paths, and everyday commuting. The device is not designed for aggressive riding conditions such as mountain biking, jumps, or steep downhill trails.
- Before each ride, inspect the device for any visible signs of wear, cracks, deformation, or malfunction. If any component appears damaged or does not operate correctly, do not use the device until it has been repaired or replaced.
- Users are encouraged to practice using the braking mechanism in a safe and controlled area before riding in traffic or crowded environments.
- If the device needs to be mounted on a different bicycle, please contact the team to ensure that the installation is performed correctly.

5. Contacts

For any questions regarding this user manual or the VeloBrake Assist device, please contact the development team at:

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